Construction Materials Lab

Hands-on training in testing and identifying essential civil engineering materials.

Official Source Declaration

This lesson is strictly based on the **Official Syllabus PDF (Revision 2026)** provided by SITTTR Kerala for Course Code **2018**.

- **Program:** Civil Engineering and related branches.
- **Authority:** State Institute of Technical Teachers' Training and Research (SITTTR).
- **Accuracy:** All experimental procedures, mapping, and assessment schemes are derived directly from the official document.

Course Dashboard

Teaching Scheme

- **L:T:P:R** — 0:0:3:0
- **Contact Hours:** 45 Hours / Semester
- **Self-Learning:** 22.5 Hours / Semester
- **Credits:** 1.5

Examination Scheme

- **CIE (Internal):** 60 Marks

- **ESE (External):** 40 Marks
- **Total Marks:** 100

How to Study this Lab Course

This is a practical-oriented course. To excel:

1. **Pre-Lab:** Read the theory of the material being tested (e.g., Bricks, Cement).
2. **Observation:** Record dimensions and weights accurately during the lab session.
3. **Calculations:** Use the standard formulas for Density, Bulking, and Silt content.
4. **Record Work:** Complete the lab record immediately after the experiment while the process is fresh.

Course Outcomes (CO)

CO No.	Course Outcome Description	Taxonomy Level
CO1	Perform basic field and laboratory tests on bricks and execute setting out of a simple building.	Apply
CO2	Identify various types of soils and flooring tiles.	Apply
CO3	Identify special processed construction materials and evaluate sand quality through various tests.	Apply
CO4	Determine the bulk density of aggregates and perform bar bending.	Apply

Syllabus Coverage

Module	Experiments	Hours
Module I	Brick Tests (Dimension, Weight, Density, Field tests), Cement Mortar, Setting out.	12
Module II	Soil identification in foundation pit, Flooring tiles identification.	6
Module III	Special processed materials, Bulking of Sand, Silt Content.	9
Module IV	Bulk Density (Fine & Coarse), Bar Bending (Hooks, Stirrups, Ties, Crank bars).	12
Final	Open Ended Experiment & Series Exams.	6

Learning Roadmap

The course starts with **Basic Units** (Bricks & Mortar), moves to **Site Investigation** (Soil & Layout), explores **Material Properties** (Sand quality), and concludes with **Structural Components** (Aggregates & Steel Reinforcement).

Module I: Bricks, Mortar & Setting Out

Instructional Hours: 12 | **Mapped CO:** CO1

Expt 1 & 2: Testing of Bricks

1. Dimension, Weight and Density

Standard size of a brick is 19cm x 9cm x 9cm. In this experiment, we measure 10 bricks to find the average.

> **Malayalam support:** □□□□□□□□□□□□□ (□□□□□□□) □□□□□□□□, □□□□, □□□□□□□□□ (Density) □□□□□□□ □□□□□□□□□□ □□□□□□□□□□□□□□□. 10 □□□□□□□□□ □□□□□□□□ □□□□□□□ □□□□□ □□□□□□□ □□□□□□□ □□□□□□□□□□□.

2. Field Tests on Bricks

- **Dropping Test:** Drop from 1m height; it should not break.
- **Striking Test:** Two bricks struck together should produce a clear metallic ringing sound.
- **Scratching Test:** Scratch with a fingernail; no impression should be left.

Expt 3 & 4: Mortar & Setting Out

Cement Mortar (1:3 or 1:6)

Preparation of a workable mix using cement and sand. 1:3 means 1 part cement to 3 parts sand by volume.

Setting Out of Building

The process of transferring the building plan from paper to the actual ground using pegs, strings, and measuring tapes.

Module II: Soil & Flooring Tiles

Instructional Hours: 6 | **Mapped CO:** CO2

Expt 5: Soil Identification in Foundation Pit

Identifying soil layers (strata) like clay, silt, sand, or gravel by visual inspection and touch in a dug pit.

> **Malayalam support:** [illegible] (Layers) [illegible].

Expt 6: Flooring Tiles Identification

Study of various tiles:

- **Vitrified Tiles:** Low porosity, very hard.
- **Ceramic Tiles:** Glazed surface, used for walls/floors.
- **Mosaic:** Marble chips embedded in cement.
- **Paving Blocks:** Used for outdoor walkways.

Module III: Sand Quality & Special Materials

Instructional Hours: 9 | **Mapped CO:** CO3

Expt 8: Bulking of Sand

Bulking is the increase in volume of sand due to surface moisture. It is maximum at 4-6% moisture content.

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% Bulking = [(V1 - V2) / V2] × 100
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Expt 9: Silt Content in Sand

Silt reduces the strength of mortar. It is determined by the decantation method in a measuring cylinder.

> **Malayalam support:** [illegible] (Silt) [illegible]. [illegible].

Module IV: Aggregates & Bar Bending

Instructional Hours: 12 | **Mapped CO:** CO4

Expt 10 & 11: Bulk Density

Determining the mass of aggregate required to fill a unit volume container.

Given: Mass of empty cylinder = 5kg, Mass with aggregate = 25kg, Volume = 0.01 m^3.
Calculation: Net Mass = 20kg. Density = 20 / 0.01 = 2000 kg/m^3.
Answer: 2000 kg/m^3.

Expt 12: Bar Bending

Practical training in bending steel reinforcement bars to specific shapes:

- **Hooks:** Standard 180° bends.
- **Stirrups:** Rectangular or square ties for beams/columns.
- **Crank Bars:** Bent-up bars used in slabs.

Student Activities for Self-Learning

Week	Activity Description	Hours
1-2	Interpreting theory concepts and finishing lab record.	3.0
3	Refer IS codes for cement specifications.	1.5
4	Identify methods of setting out buildings.	1.5
15	Identification and study of engineering problems.	1.5

Formula Bank

Density of Brick

`ρ = M / (L × B × H)`

Silt Content %

`(V_silt / V_sand) × 100`

Bulk Density

`Mass of Aggregate / Volume of Container`

Visual Library

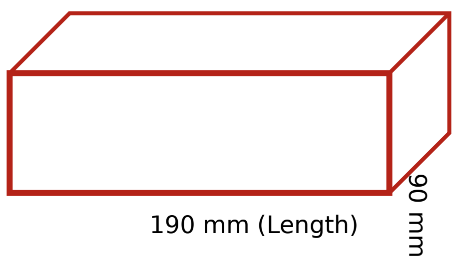

Figure 1: Standard Modular Brick Dimensions

Assessment Methodology

CIE (60 Marks)

- Attendance: 5 Marks
- Continuous Evaluation: 30 Marks
- Practical Tests (CA1 & CA2): 15 Marks
- Open-Ended Experiment: 10 Marks

ESE (40 Marks)

- Procedure/Write-up: 10 Marks
- Execution: 10 Marks
- Result/Output: 8 Marks
- Viva Voce: 8 Marks
- Record: 4 Marks

Viva Questions & Answers

What is the purpose of the 'Frog' in a brick?

The frog (indentation) serves two purposes: it forms a key for the mortar to hold the bricks together, and it can also carry the manufacturer's name.

Why does sand bulk when wet?

Surface tension of the moisture film around sand particles pushes them apart, increasing the overall volume. This effect disappears when sand is fully saturated.

What is a 'Stirrup' in bar bending?

A stirrup is a loop of reinforcing bar used to hold the main reinforcement bars together and to resist shear forces in beams.

Define Silt Content and its limit.

Silt content is the fine particle matter (less than 75 microns) in sand. Generally, it should not exceed 5% to 8% by volume for quality construction.

Practice Model Paper

(Non-official Practice Scenario)

Time: 3 Hours | Max Marks: 40

1. Write the procedure to determine the silt content of a given sand sample. (10 Marks)
2. Perform the field tests on the provided brick samples and record your observations. (10 Marks)
3. Tabulate the dimensions of 5 bricks and calculate their average density. (10 Marks)
4. Viva Voce: Explain the 1:4 mortar ratio. (10 Marks)

Quick Revision

- **Brick Size:** 19 x 9 x 9 cm (Modular).
- **Bulking:** Max at ~5% moisture.
- **Density:** Mass per unit volume (kg/m^3).
- **Setting Out:** 3-4-5 rule is commonly used for right angles.
- **Bar Bending:** Standard hook length is usually 9d or 10d.

Glossary

Mortar
: A mixture of binding material (cement), fine aggregate (sand), and water.

Strata
: Distinct layers of soil or rock found in the ground.

Vitrified
: A process where tiles are fired at high temperatures to make them glass-like and non-porous.

Decantation
: Separating a liquid from a solid by pouring off the liquid after the solid has settled.

References

- *Construction Materials* by Ghose, D. N. (Tata McGraw Hill)
- *Civil Engineering Construction Materials* by S.K. Sharma (Khanna Publishing)
- *Building Materials* by Varghese, P.C. (PHI Learning)
- IS Codes (Bureau of Indian Standards) - bis.gov.in